Exercise 01 – SQL Fundamentals :

SQL fundamentals : SELECT & Filtering

① SELECT *

FROM employees_table ;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

② SELECT DISTINCT department

FROM employees_table ;

department
IT
HR
Finance
Marketing

③ SELECT first_name,

last_name

FROM employees_table

ORDER BY salary DESC ;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ SELECT id,
first_name,
last_name,
Salary
FROM employees_table
ORDER BY Salary DESC
LIMIT 3 ;

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

⑤ SELECT id,
first_name,
last_name,
department
FROM employees_table ;
WHERE department = 'IT' ;

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	marketing
5	Eva	Naidoo	IT

(6) SELECT id,
first_name,
last_name,
department,
Salary
FROM employees_table
WHERE salary > 60000 AND department = 'Finance';

id	first_name	last_name	department	Salary
3	Cathy	Zilly	Finance	65000

(7) SELECT id,
first_name,
last_name,
department
FROM employee_table
WHERE department = 'HR' OR ~~finance~~ department = 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

(8) SELECT id,
first_name,
last_name,
department
FROM employees_table
WHERE department NOT IN ('IT');

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

(9) SELECT id,
first_name,
last_name,
department
FROM employees_table
WHERE department IN (IT, HR) OR 'Finance';

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Eva	Naidoo	IT

(10) SELECT id,
first_name,
last_name,
department,
Salary,
City
FROM employees_table
WHERE Salary > 6500 AND City = 'Johannesburg' IN ('IT');

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	7000	Johannesburg
5	Eva	Naidoo	IT	7000	Johannesburg